

Alexander Gardiner

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EDUCATION

University of California, Davis **Expected May 2028**
B.S. in Computer Science - GPA: 3.9 Davis, CA
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Computer Organization

PROFESSIONAL EXPERIENCE

Founding Software Engineer **Sep 2025 - Present**
Clove Agency – AI-Powered Marketing Intelligence Platform Santa Barbara, CA

- Expanded marketing **data coverage from 1 to 5+ pipelines**, including customer reviews and forum posts, into an AWS Lightsail and PostgreSQL RDS platform, enabling semantic search to broaden cross-channel analysis across 5+ client accounts.
- Engineered a large-scale ingestion pipeline **processing 1M+ Google Ads records** and automated **sentiment analysis across 100,000+ customer reviews** via event-driven Lambda + SQS with asynchronous LLM analysis delivering near-real-time insights for marketing teams.
- Reduced deployment time from 45 minutes to 1.5 minutes** by migrating from manual deployments to a GitHub Actions CI/CD pipeline featuring Docker-based blue-green deployments and zero-downtime releases.

Software Engineer (Contract) **Jan 2026 - Present**
Benevolent Bandwidth Foundation – Global Nonprofit | benevolentbandwidth.org | Advisors from PayPal, Capital One, Google, Amazon Davis, CA

- Built a lightweight custom YAML-driven orchestration engine** with LLM routing to specialized agents including modular tool execution and skill loading, replacing multi-week manual agent development with configuration-driven deployment.
- Improved zero-shot garment tag attribute recognition from 65% to 99%** on a 100-image client-provided validation set by replacing legacy OCR with a GCP + VLM vision pipeline, integrating into a React Native app to estimate carbon emissions.

PROJECTS

IEEE ITAIC 2025 Peer Reviewed Published Research Paper - Child vs. Adult Speech Classification - [Paper] **Aug 2024 - May 2025**
Pytorch, Neural Networks, Transformers, Impulse Response Convolution, Google Colab

- Achieved **93.5% accuracy and 93.81% F1 on child vs. adult speech classification** on 2 held-out cross-corpus test sets across classical models (logistic regression, XGBoost, neural networks) using transformer-based embeddings (HuBERT, Whisper, ECAPA-TDNN).
- Increased **training data 4x to 31,936 utterances** by unifying 6 datasets and applying impulse response convolution augmentation to normalize recording artifacts, including 2 private datasets from UCLA's Speech and Auditory Processing Lab.

Pest Spread Prediction and Notification App – Anticipate **HackDavis, May 2026**
React Native, Postgres (Supabase), FastAPI, Vercel, Expo, Gemini AI

- Built a field-level **agricultural risk platform across 6,731 Yolo County fields** using React Native, FastAPI, Gemini Vision AI, and UC IPM vector search, generating reports with BFS pest propagation and PostGIS geospatial scoring.
- Shipped an end-to-end farmer alert system on TestFlight with **proactive alerts across 63 irrigation districts**, supporting offline retry uploads and map visualization via Expo push notifications.

Cloud-Native Media Review Platform – Reel Feedback **Oct 2025 - Present**
Cloudflare, AWS Lambda, Next.js, HLS Transcoding

- Architected a scalable serverless video platform using **Cloudflare Workers, Next.js, and Cloudflare D1**, enabling low-latency playback with Google OAuth-secured access across distributed user sessions.
- Optimized processing-to-playback availability latency to ~7–20s** for short form content by engineering a HLS video processing pipeline using AWS Lambda and FFmpeg to generate 4–7 adaptive bitrate renditions (1080p–360p).

LEADERSHIP

Vice-President of Infrastructure, CodeLab Davis **Sep 2025 - Present**
CodeLab, UC Davis - Software and Design Agency Davis, CA

- Assessed 100+ applicants and conducted 10+ technical interviews for a 60+ member agency**; architected an AWS Lambda + SQS pipeline with Rekognition and Nova embeddings for semantic image search, **with sub-200ms retrieval latency using pgvector**.

Founding Software Lead **Aug 2023 - June 2025**
Octobots - First Robotics Competition (High School) Santa Barbara, CA

- Founded and led a software team of 15 engineers** to elevate team ranking from 43rd to 10th out of 300 in California within one year through autonomous trajectory adherence, YOLO object detection, and precise alignment using AprilTags and PID.

SKILLS

Languages / Frameworks: C/C++, Python, JavaScript, TypeScript, React.js, Next.js, Java, SQL, React Native, FastAPI, Node.js
Tools: Git, Docker, MongoDB, Expo, MySQL, nginx, PostgreSQL, pgvector, CI/CD, GitHub Actions, Supabase, Vercel, PostGIS, SQLite
Cloud: AWS (Lightsail, Lambda, SQS, RDS, S3, Route 53, ECR, Fargate), GCP (App Engine, AI Studio), Cloudflare (Workers, D1, R2)
AI / ML: PyTorch, TensorFlow, Neural Networks, YOLO, Dataset Augmentation, Feature Embeddings, NLP, Information Retrieval
Awards: 1st Place @ UCSB International CTF (High School Division)